



SCHOOL OF COMPUTATION,
INFORMATION AND TECHNOLOGY —
INFORMATICS

TECHNISCHE UNIVERSITÄT MÜNCHEN

Bachelor's Thesis, Master's Thesis, ... in Informatics

Thesis title

Author





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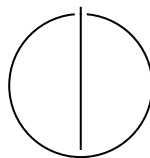
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Titel der Abschlussarbeit

Author:	Author
Examiner:	Supervisor
Supervisor:	Advisor
Submission Date:	Submission date



I confirm that this bachelor's thesis, master's thesis, ... is my own work and I have documented all sources and material used.

Munich, Submission date

Author

Acknowledgments

Abstract

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1 Introduction

1.1 Section

Citation test [Lam94].

Acronyms must be added in `main.tex` and are referenced using macros. The first occurrence is automatically replaced with the long version of the acronym, while all subsequent usages use the abbreviation.

E.g. `\ac{TUM}`, `\ac{TUM}` $\Rightarrow$ Technical University of Munich (TUM), TUM

For more details, see the documentation of the acronym package¹.

1.1.1 Subsection

See Table 1.1, Figure 1.1, Figure 1.2, Figure 1.3.

Table 1.1: An example for a simple table.

A	B	C	D
1	2	1	2
2	3	2	3

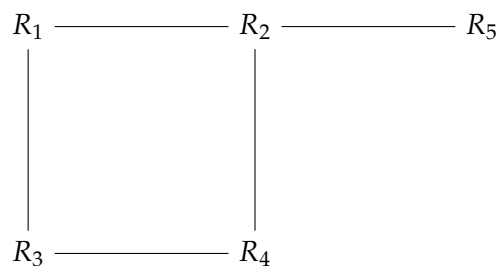


Figure 1.1: An example for a simple drawing.

¹<https://ctan.org/pkg/acronym>

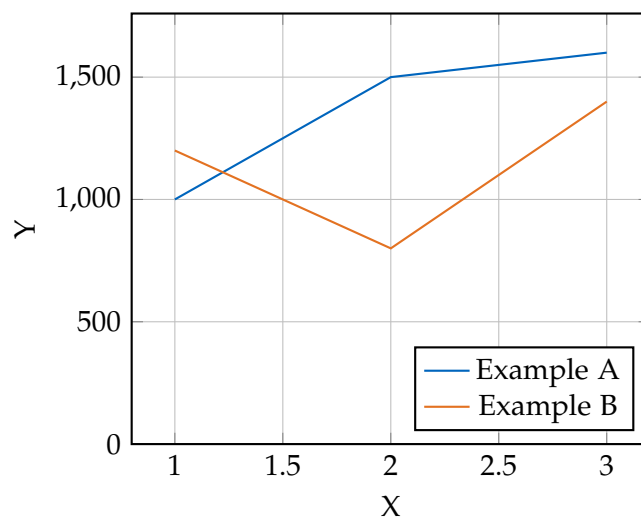


Figure 1.2: An example for a simple plot.

```
SELECT * FROM tbl WHERE tbl.str = "str"
```

Figure 1.3: An example for a source code listing.

2 Related Work

Techniques for Register allocation have focused either primarily on code quality or on compilation speed. Graph coloring based register allocation [Chaitin1981] has been shown to produce good quality code but can be slow for large functions due to the complexity of graph coloring algorithms [Appel1998]. Linear scan register allocation [Poletto1999] is a faster alternative that trades off some code quality for speed, making it suitable for just-in-time compilation scenarios.

Most(TODO all?) literature considers register allocation after having performed Instruction Selection. TODO woanders

2.1 Graph Coloring Register Allocation

Graph coloring register allocation [Chaitin1981] models the register allocation problem as a graph coloring problem, where each node represents a variable and an edge between two nodes indicates that the variables are live at the same time and thus cannot share the same register.

While graph coloring in general is a NP-complete problem [Chaitin1981], programs in SSA form have interference graphs that are chordal [Hack06], allowing for polynomial time coloring algorithms [Hack06].

Standard graph coloring approaches are dominated by the cost of building and rebuilding the interference graph after spilling [Cooper06]. By reusing parts of the interference graph after spilling, the allocation time can be reduced by around 50% [Cooper06].

Abbreviations

TUM Technical University of Munich

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Bibliography

- [Lam94] L. Lamport. *LaTeX : A Documentation Preparation System User's Guide and Reference Manual*. Addison-Wesley Professional, 1994.